Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 40)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain the locational requirements for harnessing wave power.

[2 marks]

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0 1 . 2 Describe how **one** named wave power device generates electricity.

[2 marks]

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0 1 . 3 Give **two** design problems that have delayed the development of wave power.

[2 marks]

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0 1 . 4 Compare the environmental impacts of tidal barrages, tidal lagoons and in-stream tidal turbines.

[4 marks]

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0 2 Two insecticides, X and Y, were tested on a pest insect and on a predatory ground beetle that feeds on the pest. Table 1 shows the results.

Table 1

Insecticide	LD_{50} for pest / mg kg^{-1}	LD_{50} for ground beetle / mg kg^{-1}	Half-life in soil / days	Solubility in water
X	5	6	180	Low
Y	8	150	4	High

0 2 . 1 Explain what is meant by LD_{50}.

[1 mark]

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0 2 . 2 Using Table 1, explain which insecticide is more likely to cause indirect effects on the pest population through the food chain.

[3 marks]

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0 2 . 3 Suggest **two** environmental problems that could be caused by insecticide Y that are less likely with insecticide X.

[2 marks]

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0 2 . 4 Explain why the half-life of insecticide X makes it more likely to bioaccumulate.

[2 marks]

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0 2 . 5 Give **two** reasons why a systemic insecticide may be preferred to a contact insecticide.

[2 marks]

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0 3 A factory discharges a persistent, water-soluble pollutant into a river. Table 1 shows the concentration of the pollutant measured at sites along the river on a day with normal river flow.

Table 1

Site	Distance downstream of discharge / km	Pollutant concentration / mg dm^{-3}	River flow / $m^3 s^{-1}$
A (upstream)	-2	0.0	12
B	0.5	4.8	12
C	5	3.9	15
D (below tributary)	12	1.6	36
E	30	1.4	38

0 3 . 1 Calculate the percentage decrease in pollutant concentration between site B and site E.

[2 marks]

Answer ..... %

0 3 . 2 Explain the large decrease in concentration between sites C and D.

[2 marks]

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0 3 . 3 Explain why the concentration changes very little between sites D and E.

[2 marks]

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0 3 . 4 Suggest why the pollution might be more serious if the same discharge entered a lake instead of a river.

[2 marks]

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0 3 . 5 Explain why the factory might be permitted to discharge the effluent into the sea near a strong current but not into the river.

[2 marks]

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0 4 . 1 Explain what is meant by the **embodied energy** of a vehicle and how it can be reduced.

[3 marks]

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0 4 . 2 Describe **four** principles of designing vehicles for end of life.

[4 marks]

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0 4 . 3 Explain why there is an optimum speed for fuel efficiency in a car.

[2 marks]

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0 4 . 4 Give **one** way driver behaviour or vehicle systems can reduce fuel use in congested traffic.

[1 mark]

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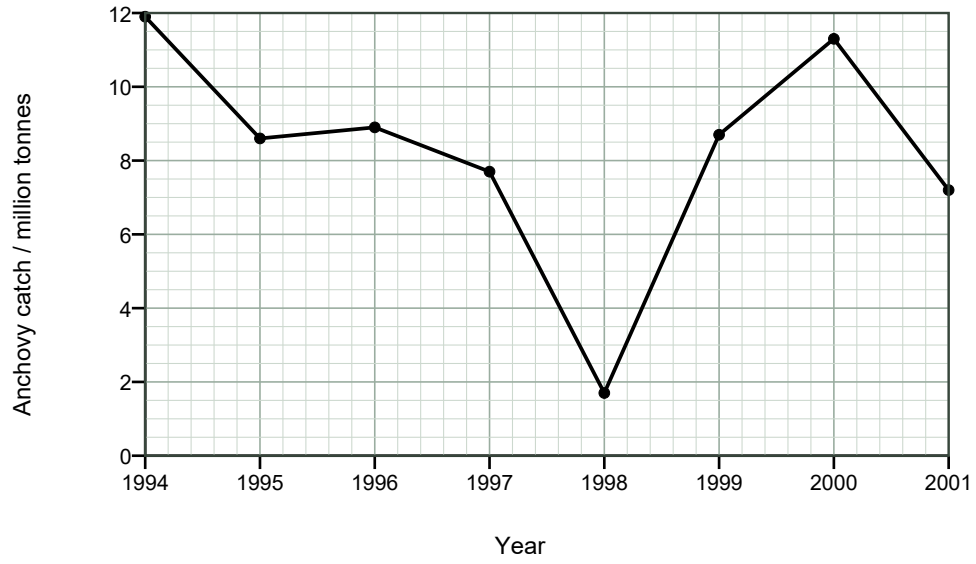
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0 5 Figure 1 shows the sea surface temperature anomaly in the eastern Pacific off Peru and the annual Peruvian anchovy catch for a series of years including a strong El Niño event in 1997-98.

Figure 1



0 5 . 1 Describe the normal (non-El Niño) pattern of winds and ocean currents in the tropical Pacific and explain why the Peruvian coast supports a rich fishery.

[4 marks]

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0 5 . 2 Explain how an El Niño event caused the collapse of the anchovy catch in 1998 shown in Figure 1.

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0 5 . 3 Describe **two** other effects of El Niño events.

[2 marks]

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0 5 . 4 Suggest why some scientists link global climate change to changes in the frequency of El Niño events, and why this link is uncertain.

[1 mark]

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0 6 . 1 Describe how soil texture is determined using a stack of sieves.

[3 marks]

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0 6 . 2 Explain why a soil auger gives a more representative sample than a trowel.

[2 marks]

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0 6 . 3 Explain why the organic matter content will be overestimated if a soil sample is heated above 550 °C.

[2 marks]

0 6 . 4 Describe how a turbidimeter measures water turbidity and explain how quantitative results are obtained.

[3 marks]

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0 7 In the 1950s a chemical factory at Minamata, Japan, discharged inorganic mercury compounds into a shallow bay. Thousands of local people who ate fish and shellfish from the bay suffered severe neurological damage and many babies were born with birth abnormalities.

0 7 . 1 Explain how the low-toxicity inorganic mercury discharged into the bay came to cause such severe poisoning.

[4 marks]

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0 7 . 2 Explain why methyl mercury caused both neurological damage in adults and birth abnormalities.

[3 marks]

0 7 . 3 Compare the toxicity of the three chemical forms of mercury.

[3 marks]

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0 8 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 8 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 8 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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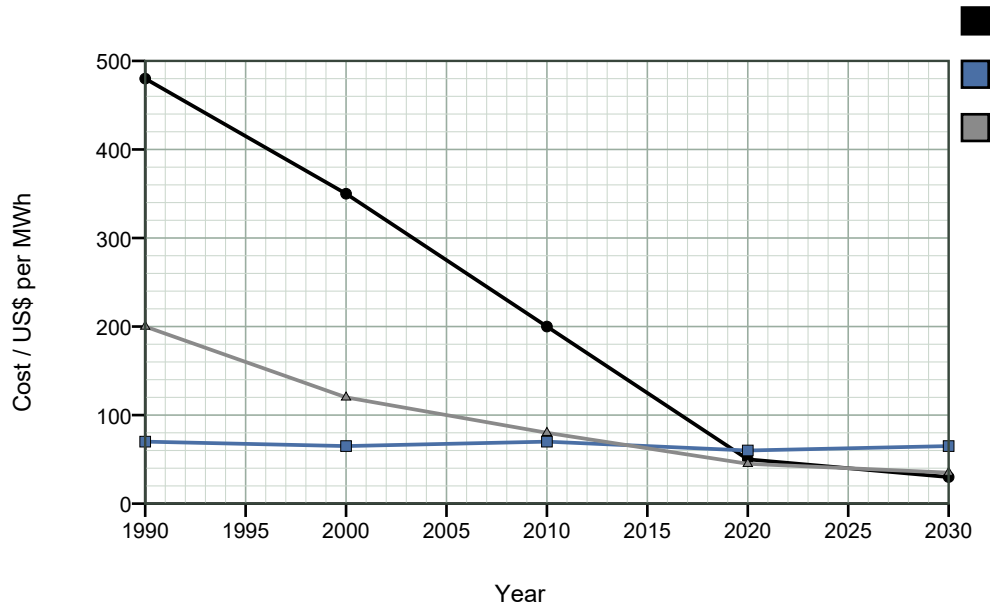
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0 9 Figure 1 shows the estimated cost of generating electricity from three sources between 1990 and 2020, and projections to 2030.

Figure 1



0 9 . 1 Describe the trends shown in Figure 1.

[3 marks]

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0 9 . 2 Explain why new energy technologies are expensive when first introduced and why their costs then fall.

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0 9 . 3 Explain why a renewable energy project may still be at a financial disadvantage compared with a gas-fired power station of the same total cost.

[2 marks]

0 9 . 4 Explain why the price paid for fossil fuel energy does not reflect its full cost.

[2 marks]

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1 0 . 1 Describe how human activities have altered the natural equilibria of the carbon and nitrogen cycles, and evaluate the strategies that could counteract these changes.

[9 marks]

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1 0 . 2 State what is meant by the **residence time** of a substance in a reservoir.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Discuss how the properties of solar power and hydroelectric power affect their use, and evaluate how new technologies could increase their contribution to energy supplies.

[25 marks]

OR

11 . 2 Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS